

1. LOCKOUT / TAGOUT (LOTO) PROCEDURE

Before any service work, the authorized technician must follow the company-mandated lockout/tagout procedure as required by OSHA 29 CFR 1910.147 and ISO 9001 section 7.1.3. Failure to comply may result in severe injury or death.

Step 1. NOTIFY: Inform all affected operators that the machine is being taken out of service for maintenance. Post the maintenance notice on the HMI screen.

Step 2. IDENTIFY ENERGY SOURCES: Identify all energy sources for the specific machine being serviced:

- Electrical: main disconnect switch, control transformer, DC bus capacitors
- Pneumatic: compressed air supply line, accumulator tanks
- Hydraulic: main pump, accumulators, pressurized lines
- Mechanical: flywheel kinetic energy, spring-loaded mechanisms, counterbalance cylinders, suspended tooling

Step 3. SHUTDOWN: Shut down the machine using the normal stop procedure. Allow the machine to come to a complete stop. For flywheel machines, this may take up to 2 minutes.

Step 4. ISOLATE: Disconnect or isolate each energy source. Lock the main electrical disconnect switch in the OFF position using a personal padlock. Each technician working on the machine must apply their own lock.

Step 5. DISSIPATE STORED ENERGY: Bleed hydraulic pressure by opening the pressure bleed valve. Discharge DC bus capacitors by shorting across the bus bars with a grounded discharge rod (rated for 1000V DC). Lower all suspended tooling or blocks. Release spring tension.

Step 6. VERIFY ZERO ENERGY: Attempt to start the machine using the normal start button. The machine must not start. Verify that all motion has stopped and all pressure gauges read zero.

Step 7. TAG: Apply a tag identifying the technician, the date, the expected completion time, and a description of the work being performed.

WARNING: Never attempt to service a machine without completing all seven steps. Always verify zero energy state before touching any component.

2. CROSS-MACHINE ERROR CODE REFERENCE

This facility operates multiple machine types. The same error code may have DIFFERENT meanings on different machines. Always confirm the machine model before diagnosing any fault.

E101 ? CRITICAL: Different on every machine:

- RoboInject-300: Winding overtemperature (see RoboInject manual §4.2, p.214)
- Press-2000: Low hydraulic pressure (see Press-2000 manual §3.1, p.92)
- Press-2001: Drive chain binding (see Press-2001 manual §7.1, p.311)

E204 ? Also varies by machine:

- RoboInject-300: Servo stall detected (§8.3, p.312)
- Press-2000: High pressure exceeded (§5.2, p.138)
- Press-2001: Flywheel overspeed (§7.2, p.315)

E302 ? Varies:

- RoboInject-300: Zone 1 heater overcurrent (§9.1, p.378)

Common codes (same meaning across all machines):

- E011: Safety loop open ? one or more safety switches are not closed
- E099: Emergency stop activated ? one or more E-stop buttons are pressed
- A107: Low lubrication oil level ? central lubrication system needs refill

IMPORTANT: If you are unsure which machine the error code refers to, identify the machine by its serial number prefix before proceeding:

- SM-: RoboInject-300 (injection molding)
- HY- or HP-: Press-2000 (hydraulic press)
- MP-: Press-2001 (mechanical press)

3. GENERAL SAFETY RULES

3.1 PERSONAL PROTECTIVE EQUIPMENT (PPE)

All personnel in the production area must wear:

- Hearing protection: earplugs or earmuffs rated ?25 dB NRR
- Safety glasses with side shields (ANSI Z87.1 compliant)
- Steel-toed boots (ASTM F2413)
- Close-fitting clothing (no loose sleeves, ties, or jewelry)
- Gloves are prohibited around rotating machinery

3.2 HOUSEKEEPING

Keep the area around each machine clear of debris, oil spills, and unused tools. Any hydraulic or oil spill exceeding 0.5 L must be reported to the shift supervisor and cleaned using the spill kit located at each bay entrance. Use absorbent pads for oil and designate a waste disposal container.

3.3 EMERGENCY PROCEDURES

In the event of an injury:

1. Press the nearest E-stop immediately.
2. Call for assistance using the emergency phone (ext. 111) or the nearest emergency pull station.
3. Provide first aid if trained and safe to do so.
4. Do not attempt to move a person with suspected spinal injury.

First-aid kits are located at each bay entrance. Fire extinguishers (Type ABC, 10 lb) are mounted at 20-meter intervals along the building columns. Automated external defibrillators (AEDs) are located at the main break room and the shipping/receiving office.

3.4 REPORTING NEAR MISSES

Any near-miss incident (e.g., a tool dropped inside a machine guard, a hydraulic line that burst while the machine was idle, a safety switch that failed to actuate during testing) must be reported to the safety department within 24 hours using form SAF-101. Near-miss reporting is a key component of ISO 9001 continuous improvement and is not disciplinary.